

FANR 3800: Week 2 Lecture Plan

Arc A · Telling the spatial story

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1 Week 2 - Telling the Spatial Story

Arc A: Build a base you can trust. Two 50-minute lectures (Mon & Wed). Map communication is taught early, deliberately, so students can present every subsequent analysis credibly. The theme: a map is an argument, and design determines whether the argument is believed.

Notation: ArcGIS Pro tools appear as ALL-CAPS; methods and concepts as Proper-Case italics.

1.1 Monday: *Symbology: Making Attributes Visible*

Learning objectives. Students can (1) match a *Symbology* method to a data type, (2) explain how a symbology choice changes the story a map tells, and (3) avoid the common misreadings caused by poor symbol choices.

Session outline (~50 min).

- (0–10) *Why symbology is analysis, not decoration.* Same data, two symbol schemes, two different conclusions. Set up the responsibility that comes with representation.

- (10–30) *The main methods.* *Single Symbol* (location only); *Categorized / Unique Values* (qualitative classes, species, ownership, cover type); *Graduated* (quantitative, age, basal area, density) and the effect of classification method and class breaks. Tie each to a data type from Week 1's attribute tables.
- (30–45) *Choosing well.* Color for categories vs. sequences; number of classes; when a choropleth misleads (raw counts vs. normalized rates); accessibility of color choices.
- (45–50) *Setup for Wednesday.* A well-symbolized layer still isn't a finished map, preview *Map Layout*.

Framing to emphasize. Landowners and stakeholders act on the map, not the attribute table. *Symbology* is where the data becomes a decision.

Tools introduced: none, the SYMBOLOGY pane is a ribbon/pane workflow, not a geoprocessing tool. *Methods/concepts introduced:* *Symbology* (*Categorized, Graduated, Unique Values*), *Cartographic Communication*.

1.2 Wednesday: *Map Layout*: The Anatomy of a Credible Map

Learning objectives. Students can (1) list the essential map elements and justify each, (2) apply visual hierarchy so the main message reads first, and (3) export a publication-ready map with complete source information.

Session outline (~50 min).

- (0–15) *The essential elements.* Title, legend, scale bar, north arrow, and, emphasized, data source and coordinate system metadata. Why each exists and what a missing element costs the reader's trust.
- (15–30) *Composition and hierarchy.* Guiding the eye: figure-ground, balance, what to enlarge and what to subordinate; the difference between a map made for a report page and one made for a wall.
- (30–42) *Credibility and honesty.* Scale and audience; how framing, extent, and class breaks can distort; the professional obligation to represent uncertainty and cite sources.
- (42–50) *Export and delivery.* Formats and resolution for print vs. screen via EXPORT LAYOUT; naming and file organization consistent with the course conventions.

Framing to emphasize. Good design makes the spatial story clear; poor design obscures it and undermines the analyst's credibility. This is the deliverable skill every arc ends on.

Connections. Every arc culminates in a map or summary; the elements introduced here are expected on all lab deliverables and exams.

Tools introduced: EXPORT LAYOUT (the LAYOUT view itself is a ribbon/pane workflow). *Methods/concepts introduced:* *Map Layout, Cartographic Communication*.